When Post-Editing Helps and When It Harms: Evaluating LLM Post-Editing of NMT for Tagalog and Spanish

Norris Research Project — Working Draft
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***Abstract*—Large language models (LLMs) can post-edit neural machine translation (NMT) drafts, but an editor that repairs errors can also change names, numbers, or relations that the draft already rendered correctly. We evaluate this trade-off for English–Tagalog and English–Spanish in both directions on the held-out FLORES+ devtest partition (1,012 sentences per direction). A sentence-split NLLB-200 3.3B system supplies the draft; Llama 3 8B Instruct and GPT-4o Mini edit it under three prompting strategies: zero-shot, one retrieved development exemplar, and a single-call reason-then-edit format. All prompts, output contracts, and analyses were frozen on development data before the test partition was opened, and every planned output, including contract failures that fall back to the baseline, is evaluated. [PENDING: two to three sentences with the main held-out findings for RQ1–RQ4: largest/smallest COMET changes vs. the baseline with Holm-adjusted significance, strategy contrasts, protect-pool over-editing rates, reason-then-edit failure rates]. The results show that average quality, preservation of strong drafts, and output validity must be measured separately when deciding whether to deploy an LLM post-editor.**

***Index Terms*—automatic post-editing, machine translation, Tagalog, Spanish, over-editing, LLM evaluation**

I. Introduction

An automatic post-editor must decide both what to correct and what to leave alone. A fluent rewrite can repair an omission, but it can also alter a name, number, or source relation that the machine translation (MT) draft already expressed correctly. When drafts are strong, mean quality changes therefore give an incomplete account of editing behaviour. LLMs can improve neural MT output, yet they also produce hallucinated edits [1].

We study this tension for English–Tagalog and English–Spanish in both directions. A fixed NLLB-200 3.3B system [2] produces sentence-split drafts, and two instruction-following editors, open-weight Llama 3 8B Instruct and commercial GPT-4o Mini, receive the source and the draft. The two language pairs provide a limited resource contrast; language, direction, domain, and baseline quality remain intertwined, so we report pair and direction differences descriptively rather than as causal resource effects. We ask: **RQ1:** How do editing gains and harms differ between the Tagalog and Spanish pairs, conditional on baseline quality? **RQ2:** How do the three prompting strategies compare within each model and direction? **RQ3:** When do editors change already strong drafts and degrade their measured quality or source fidelity? **RQ4:** How do effects differ between the two directions of a language pair?

This is an empirical evaluation, not a new post-editing architecture. Anchoring an LLM editor on a frozen NMT draft and selecting in-context exemplars by semantic similarity follow established prior work [3], [4], and we do not claim either as a contribution. Our contributions are:

- a held-out evaluation in a new empirical setting (four Tagalog/Spanish directions) against a stronger, sentence-split NLLB-200 baseline, with the protocol frozen on development data before the test partition was opened;
- a two-editor comparison (open-weight and commercial) across three prompting strategies, with paired confirmatory tests and Holm multiplicity control;
- explicit failure accounting: every planned output is retained, and output-contract failures are reported as outcomes rather than discarded;
- an over-editing and preservation analysis of strong drafts that separates metric-defined harm from heuristic preservation screens.

II. Related Work and Contribution Boundary

LLM post-editing. Raunak et al. [1] formalize direct translation post-editing with GPT-4 and report both quality gains and hallucinated edits. Ki and Carpuat [5] guide LLM post-editing with external error annotations, and xTower [6] trains a model to explain and correct translation errors, with expert assessment of its explanations. Our reason-then-edit condition differs: an unmodified editor writes its own brief diagnosis in the same call, and we treat that text as an observable output, not as a verified explanation. LangMark [7] provides human post-edit triplets; FLORES+ references are independent translations, so our reference-based gains are quality proxies and do not measure reduced human editing effort.

Exemplar selection. Vilar et al. [4] analyse prompting and example selection for translation. We retrieve one development exemplar with LaBSE [8]; the similarity is semantic, and one deterministic exemplar is a reproducibility choice, not a claim of optimality.

Relationship to our earlier ICAI study. Ahmadnia et al. [3] study English-to-German test-time domain adaptation with a frozen OPUS-MT draft, GPT-4 Turbo, multilingual Sentence-BERT top-three exemplars, and an optional second terminology-verification call. Frozen-NMT anchoring and semantic exemplar retrieval are shared with that work and are cited here as established methodology. The present study changes the empirical setting (four Tagalog/Spanish directions, FLORES+), the baseline (sentence-split NLLB-200 3.3B), and the editors (one open-weight and one commercial), uses a single-call reason-then-edit instead of a second verification call, and adds explicit failure and over-editing analysis. Because data, models, and metrics differ, we make no numerical comparison across the two papers.

Evaluation. We use COMET-22 [9], [10] as the primary metric, with BLEU [11] and chrF++ [12], [13] reported with full signatures [14]. No scalar metric replaces source inspection, so preservation screens are reported alongside quality.

III. Method

NMT baseline (S0). Drafts come from facebook/nllb-200-3.3B (FP16; four beams, no sampling, early stopping, length penalty 1.0, maximum length 512; inputs checked rather than truncated). The primary baseline segments each source with pysbd 0.3.4 (character spans, `clean=False`; English rules for Tagalog as a documented heuristic, with title, clock, *atbp.*, and parenthetical safeguards), translates multi-sentence items fragment by fragment, and joins the fragments with one space; single-sentence items reuse the unsplit hypothesis. Segmentation is lossless by construction. The unsplit system is kept as a diagnostic, outside the confirmatory tests.

Editors. We use the original Meta-Llama-3-8B-Instruct checkpoint (BF16, vLLM 0.11.2, pinned revision) [15] and the dated gpt-4o-mini-2024-07-18 snapshot [16]. Both decode with temperature 0, top-p 1, seed 42, and a 1,024-token output cap. Every request carries the same system instruction: the source is authoritative; source-supported names, numbers, dates, and meaning must be preserved; stylistic rewriting is discouraged; and an accurate, natural, grammatical draft must be returned unchanged. Source and draft are passed as delimited data.

Strategies. *Zero-shot* returns one JSON field, `edited_translation`. *Retrieved one-shot* adds one solved development example (its source, NLLB draft, and reference) before the query. *Reason-then-edit* returns a free-text `diagnosis` followed by `edited_translation` in a single call; only the translation is scored. The current query's reference never enters any prompt.

Retrieval. LaBSE [8] embeds all 997 development sources per source language (normalized, 768 dimensions); an exact FAISS inner-product index returns the most similar development item, excluding exact duplicates of the query source and breaking ties by the smaller sentence ID.

Output contract and fallback. Structured decoding (vLLM; OpenAI strict JSON schema) constrains the response shape, and a strict validator then requires valid JSON without duplicate keys, the exact field set and order, non-empty fields, and a normal stop. Otherwise, the raw response is retained and the system output is the unchanged baseline draft:

$$\hat{y}_i = \begin{cases} e_i, & \text{if the response passes the contract,} \\ b_i, & \text{otherwise.} \end{cases} \tag{1}$$

Failures are neither repaired nor selectively regenerated, so every evaluated condition is a deployable policy that includes its own failure behaviour. Passing the contract does not imply a correct translation.

IV. Experimental Protocol

Data. We use FLORES+ [2], [17] at a pinned revision: 997 `dev` and 1,012 `devtest` sentences per direction. The dataset's `fil_Latn` configuration maps to NLLB's `tgl_Latn`. Spanish is the benchmark's standard variety; we make no claim about heritage, dialectal, code-switched, or Los Angeles community Spanish. All prompt development, retrieval, and pilots used `dev` only.

Development pilot and protocol revision (development evidence). On the same 100 development IDs in all four directions (2,400 outcomes), the measured first contract asked reason-then-edit for one to six unique error categories with short notes. Llama fell back to the baseline on 227/400 such outputs (GPT-4o Mini: 8/400), almost entirely because of the diagnostic validators (e.g., repeated categories), not because of JSON parsing (399/400 parsed). Because a condition that mostly returns the baseline is not a clean strategy comparison, the principal investigator requested the minimal contract described in Section III, which we fixed together with a reliability gate (at most 20/400 fallbacks per model and 10/100 per direction) before rerunning the affected development conditions. Under the revised contract, neither editor fell back on any of its 400 development outputs (Llama 0/400, GPT-4o Mini 0/400), so the gate passed. Pilot quality estimates, which were descriptive and sensitive to single items, are reported in the repository and are not used as findings.

Freeze. Prompts, schemas, model revisions, decoding, retrieval policy, output contract, statistical families, thresholds, and code hashes were recorded in a freeze manifest before `devtest` was opened. Because the original split-generation code was not recovered, the splitter was reconstructed from the recorded specification and accepted after it reproduced all 2,991 saved development segmentations exactly; regenerated fragment translations matched 412/421 saved multi-sentence development items, the residual being FP16 beam-search numerics. On `devtest`, single-sentence items reuse the checksum-locked unsplit drafts, and every editor post-edits the same frozen S0 drafts. The matrix has 24 post-editing conditions (2 editors $\times$ 3 strategies $\times$ 4 directions) of 1,012 items each (24,288 outputs) plus the baseline.

Metrics and tests. COMET-22 (Unbabel/wmt22-comet-da, pinned checkpoint) is primary and reported as native score $\times 100$. BLEU (13a, mixed case, exponential smoothing) and chrF++ (character order 6, word order 2, β=2) are descriptive. For a comparison, d_i is the native segment COMET difference on the same sentence ID; we report $100\bar{d}$, a paired percentile

95% interval from 10,000 resamples, and a two-sided centred paired-bootstrap p-value, $p = (1 + \#\{|\bar{d}_0^*| \geq |\bar{d}|\})/(B + 1)$. Holm correction is applied within two prespecified families of 24 tests: each condition against S0 (F1), and one-shot−zero-shot, reason−zero-shot, and reason−one-shot within each editor and direction (F2). Non-significant differences are inconclusive; an interval inside ± 0.5 points is flagged as practically small, not as equivalence. Pair and direction differences (RQ1, RQ4) are described from these estimates without separate tests.

Harm and over-editing. A segment improves if $d_i > 0.02$ (two reported points), is harmed if $d_i < -0.02$, and otherwise lies within the band. The protect pool contains items whose S0 COMET is at least 0.90 and the improve pool items below 0.80, both fixed on S0 before editing. A protect-pool over-edit is a textual change (NFC, outer whitespace removed) with $d_i < -0.02$, reported over all protect items, with sensitivity at 0.01 and 0.05. Digit, capitalized-token, language-identification (langdetect), length-ratio, and sentence-count screens are heuristics, not gold error labels.

V. Held-out Results

Table I reports every condition on `devtest`. All 24 conditions contain the planned 1,012 items per direction; baseline fallbacks remain in the population.

Quality against the baseline (RQ1, RQ4). **[PENDING: per pair and direction: which conditions change COMET relative to S0 after Holm correction (F1: number significant of 24), the size of the largest gains and losses with intervals, and how the En→Tl/Tl→En pattern compares with En→Es/Es→En; pool-stratified changes (improve vs. protect pool) for the RQ1 conditional-on-quality part. Describe pair/direction differences from estimates only; do not infer them from significance patterns]**.

Strategies (RQ2). **[PENDING: F2 results: one-shot vs. zero-shot, reason vs. zero-shot, reason vs. one-shot per editor and direction, with Holm-adjusted outcomes; state inconclusive contrasts as inconclusive]**.

Output validity. **[PENDING: fallback counts per editor and strategy (zero-shot/one-shot vs. reason-then-edit v3), primary failure classes, and whether held-out failure rates match the development gate]**.

Over-editing of strong drafts (RQ3). **[PENDING: protect-pool over-edit counts/denominators per condition (over-editing table, generated by Phase 5), sensitivity at 0.01/0.05, edit rates, and the relation between edit rate and harm (band figure, generated by Phase 5)]**.

VI. Discussion and Failure Analysis

Validity is part of the system. Each condition is evaluated as deployed, including its fallback behaviour. Filtering to accepted responses would select a model-dependent subset and overstate editing benefits, so failure rates are reported alongside quality. The development pilot showed why: a strict diagnostic contract turned most Llama reason-then-edit outputs into the unchanged baseline, which is not restraint. **[PENDING: held-out failure behaviour of the minimal contract, and what remains after controlling for it]**.

Average quality and fidelity can disagree. During development, source-based inspection found an edit that changed a source-backed area figure while COMET rose, a schema-valid response that replaced a name with an unsupported one, and a source-consistent paraphrase that lost over 20 COMET points. These development cases motivate the preservation screens; they are illustrations, not population estimates. **[PENDING: held-out preservation screen counts relative to the baseline (digits, capitalized tokens, language ID, length ratio, sentence counts) and two or three source-reviewed held-out cases]**.

VII. Limitations and Ethical Considerations

Two language pairs cannot identify a causal effect of resource level: language, direction, domain, and baseline quality are confounded. FLORES+ sources are largely English-original, so reverse directions involve translated source text. References are independent translations rather than minimal post-edits, so neither editing effort nor productivity is measured, and no bilingual human evaluation was conducted; COMET and the preservation screens are imperfect proxies. One open-weight checkpoint and one dated commercial snapshot do not represent all LLMs, pretraining exposure to the public benchmark cannot be ruled out, and deterministic settings do not guarantee identical remote behaviour. Fragment-level translation can lose cross-sentence context.

The study uses public benchmark text and automatic evaluation only. Results do not establish suitability for medical, legal, or other consequential translation, and standard-Spanish results should not be generalized to community or heritage varieties. LLM diagnoses can state unsupported errors confidently and are not expert explanations. Code, prompts, frozen configuration, and numeric results are released; benchmark text and generations are not redistributed. The GPT-4o Mini runs cost about **[PENDING: US$ total from Phase 4 accounting]** at list token rates. **[PENDING: venue-required AI-assistance and funding statements, to be confirmed by the PI]**

VIII. Conclusion

We evaluated LLM post-editing of a strong, sentence-split NLLB-200 baseline for English–Tagalog and English–Spanish in both directions, with two editors, three prompting strategies, a protocol frozen before test access, and explicit accounting for failures and over-editing. **[PENDING: three sentences answering RQ1–RQ4 from the held-out evidence]**. Average gains, preservation of strong drafts, and output validity are separate properties, and an LLM post-editor should be assessed on all three before deployment. Future work should add bilingual human assessment, more language pairs, and document-level context.

TABLE I

HELD-OUT FLORES+ DEVTEST RESULTS (1,012 ITEMS PER DIRECTION; ALL PLANNED OUTPUTS, INCLUDING BASELINE FALLBACK). **[PENDING: PLACEHOLDER LAYOUT — THIS TABLE IS GENERATED BY PHASE 5 INTO GENERATED/MAIN_RESULTS_MATRIX.TEX]**

		En→Tl			Tl→En			En→Es			Es→En		
Editor	Strategy	C	Δ [95% CI]	FB	C	Δ [95% CI]	FB	C	Δ [95% CI]	FB	C	Δ [95% CI]	FB
S0 split NLLB		00.00	–	–	00.00	–	–	00.00	–	–	00.00	–	–
GPT-4o Mini	Zero	00.00	+0.00 [+0.00, +0.00]	0	00.00	+0.00 [+0.00, +0.00]	0	00.00	+0.00 [+0.00, +0.00]	0	00.00	+0.00 [+0.00, +0.00]	0
	One	00.00	+0.00 [+0.00, +0.00]	0	00.00	+0.00 [+0.00, +0.00]	0	00.00	+0.00 [+0.00, +0.00]	0	00.00	+0.00 [+0.00, +0.00]	0
	Reason	00.00	+0.00 [+0.00, +0.00]	0	00.00	+0.00 [+0.00, +0.00]	0	00.00	+0.00 [+0.00, +0.00]	0	00.00	+0.00 [+0.00, +0.00]	0
Llama 3 8B	Zero	00.00	+0.00 [+0.00, +0.00]	0	00.00	+0.00 [+0.00, +0.00]	0	00.00	+0.00 [+0.00, +0.00]	0	00.00	+0.00 [+0.00, +0.00]	0
	One	00.00	+0.00 [+0.00, +0.00]	0	00.00	+0.00 [+0.00, +0.00]	0	00.00	+0.00 [+0.00, +0.00]	0	00.00	+0.00 [+0.00, +0.00]	0
	Reason	00.00	+0.00 [+0.00, +0.00]	0	00.00	+0.00 [+0.00, +0.00]	0	00.00	+0.00 [+0.00, +0.00]	0	00.00	+0.00 [+0.00, +0.00]	0

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